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U.S. energy product supply elasticities: A survey and application to the U.S. oil market

Carol Dahl *, Thomas E. Duggan

Division of Economics and Business, Mineral Economics Department, Colorado School of Mines, Golden, CO 80401, USA

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Abstract

We survey studies of simple energy supply models to find the most promising technique for developing supply elasticities in the U.S. crude oil market. The two dozen studies located include direct estimates of energy supply elasticities or cost studies from which supply or reserve elasticities can be inferred. We include all available studies for all forms of energy both primary and secondary. We find direct estimates of oil supply to obtain weak results unless depletion and price expectations are included. Oil product supply elasticities vary widely across studies but appear to be elastic. Studies that estimate reserve price elasticities by computing reserve costs appear to be the most promising for estimating reserve elasticities for fossil fuel supply. Hence we apply this technique to US oil reserves and find a reserve elasticity of 1.27.

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1. Introduction

Energy, which provides heat, light, transportation, and power, comes in a variety of forms. Coal and natural gas are burned directly or are used indirectly for

* Corresponding author.

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electricity; hydro power and uranium are only used indirectly for electricity; while oil is processed into a number of secondary products, the bulk of which are used for transportation fuels in the U.S.

All these different forms of energy are supplied under very different conditions, and, because of different market and cost structures for the various products, we would expect their elasticities of supply to differ as well. Such energy product supply elasticities, if available, are convenient for summarizing the behavior of energy producers, for forecasting, and for policy analysis. Therefore, we consider the performance of models of U.S. energy product supply that contain direct estimates of supply elasticities or that contain cost estimates from which supply elasticities can be inferred. [1] We include all available energy supply studies including primary energy products – coal, oil, natural gas, and uranium – as well as the secondary products, electricity and individual oil products.

We begin our survey with a discussion of direct energy supply estimation in Section 2. We consider summaries of elasticities using estimates of cost in Section 3. We apply what we feel is the most fruitful approach to the US oil market to develop a reserve oil supply function in Section 4. We summarize the results of the study in Section 5.

2. Estimating energy supply directly

The simplest approach to modeling an energy product is to estimate a supply equation directly with quantity supplied a function of price and other relevant variables such as reserves or cumulative production. For such a function to exist, competitive markets are required in which producers statically optimize, as implied by arguments in Adelman (1990).

Variants of this equation have been estimated in a single equation or simultaneous systems approach in Griffin (1985), Dahl and Yücel (1991), Kandel (1990), Jones (1990), MacAvoy (1982), and Hogan (1989) oil, Labys et al. (1979) for coal, Green (1988) for residual fuel oil, and Dahl (1981) and Dahl and Laumas (1981) for gasoline. Estimated energy product or reserve elasticities for primary energy products are contained in Table 1 and those for secondary energy products are contained in Table 2. Their model type is designated as Supply.

Four of these papers have estimated simple oil supply equations while trying to study market structure. Griffin (1985) found a negative and statistically significant (statistically significant throughout this paper will indicate significant at or below the 5% level) supply elasticity of −0.05 for the US when the log of output was regressed on the log of price for data from 1971–82. Dahl and Yücel (1991) found

[1] For oil and gas supply surveys that include exploration models see Dahl and Duggan (1996), Clark et al. (1981), and Walls (1992).

price elasticity to be equal to −0.08 but statistically insignificant when the log of production was regressed on the logs of price, production cost, wells drilled, US income, and world oil production. Kandel (1990) found a negative but statistically insignificant coefficient on price when output was regressed on lagged output and price using data for 1958 to 1987 but no elasticities were reported. Jones (1990) found a negative supply elasticity when the log of crude production was regressed on the log of total OPEC output and the log of crude oil prices for data from 1971–1988.

None of these simple models performed well according to a priori expectations. The failure to take into account the simultaneous determination of supply and demand, the depletion effect, dynamic optimization, price controls, and technological change may be contributing to poor model performance.

To investigate simultaneous systems bias, MacAvoy (1982) used two methods to estimate the elasticity of supply of crude oil from 1955–1973. First, he applied OLS regression techniques to estimate the quantity supplied as a function of price, reserves, and lagged supply. None of the coefficients were found to be statistically significant. He then estimated a similar supply equation jointly with demand using 2SLS indicated in the Tables as SupplySys. In this new equation, he found that reserves and the lagged endogenous variables were statistically significant but the price elasticity of crude oil in the United States was not.

His attempts to estimate the supply of reserve discoveries as a function of lagged discoveries, lagged reserves, and lagged price were no more successful. For two subperiods of the sample, from 1958 to 1968 and from 1969 to 1978, he found negative reserve price elasticities. Furthermore, both of these time periods are rather short, particularly the latter, and they offer few degrees of freedom from which to make inferences.

In estimating both oil production and reserve additions, MacAvoy's model may be too simple. Although he accounted for simultaneity, he ignored price controls, price expectations, depletion, and technical change.

Hogan (1989) estimated a simple econometric supply equation for US oil. He regressed the log of oil production on the log of average oil prices for the last six years, a time trend, and the log of lagged production on data for 1966–1987. His results with t statistics given in parentheses are

$$\ln Q = 1.31 + \underset{(3.203)}{0.088} \ln(\Sigma P_{t-i})/6 - \underset{(-4.1348)}{0.0079}\, time + \underset{(8.9618)}{0.8479} \ln Q_{t-1}.$$

His long-run price elasticity of supply is 0.58 and all else equal, production falls over time. His results appear to be more reasonable, with the lagged prices perhaps picking up expectations or smoothing out some of the randomness, while the time trend may be picking up a depletion effect or technical change or some combination. In this instance, the negative coefficient on time suggests that the depletion effect outweighed technical improvements during this time period.

Simple supply models have also been used to analyze petroleum product

Table 1
Primary energy product or reserve supply elasticities for the United States [a]

Reference	Model type	Sample	Output or reserve price elasticity		
			Short run		Long run
Oil					
Griffin (1985)	Supply	71–82 T		−0.05	
Dahl and Yücel (1991)	Supply	71–87 T		−0.08	
Kandel (1990)	Supply	58–87 T		< 0	
Jones (1990)	Supply	71–88 T		−0.24	
MacAvoy (1982)	Supply	55–73 T		ns	
MacAvoy (1982)	SupplySys	65–73 T		ns	
Hogan (1989)	SupplyDep	66–87 T	0.09		0.58
Griffin and Jones (1986) *	Cost	83 Wells TX C		−1.39	
Oil reserves					
MacAvoy (1982)	Supply	58–68 T		−0.15	
MacAvoy (1982)	Supply	69–78 T		−0.45	
Coal					
Labys et al. (1979)	SupplySteam	55–73 T	0.61		0.05
Labys et al. (1979)	SupplyTotal	55–73 T	0.05		0.11
Labys et al. (1979)	SupplyUground	55–73 T	0.67		1.31
Labys et al. (1979)	SupplySurface	55–73 T	0.07		1.00
Zimmerman (1977) *	Cost	5 States (OH, IL, PA, KY, WV)		2.03	
Zimmerman (1981) *	Cost	Mine data, Around 1973		3.58	
Coal reserves					
Lin (1978) *	Cost	N.App. 0% backfill		0.77	
Lin (1978) *	Cost	N.App. 50% backfill		0.41	
Lin (1978) *	Cost	N.App. 100% backfill		0.79	
Lin (1978) *	Cost	C.App. 0% backfill		7.56	
Lin (1978) *	Cost	C.App. 50% backfill		7.47	
Lin (1978) *	Cost	C.App. 100% backfill		7.90	
Lin (1978) *	Cost	S.App. 0% backfill		0.79	
Lin (1978) *	Cost	S.App. 50% backfill		0.69	
Lin (1978) *	Cost	S.App. 100% backfill		0.81	
Natural gas					
Barret (1992)	Supply	60–90 T		0.014	
Barret (1992)	SupplyDiseq	60–90 T		0.02 to 0.15	
Natural gas from tight sands					
Chermak and Patrick (1995) *	Cost	88–90 CT Wells		−1.05 to −1.92	

Table 1 (continued)

Reference	Model type	Sample	Output or reserve price elasticity	
			Short run	Long run
Natural gas reserves				
Dahl (1992) *	Cost	86–89 CT	0.41	
Uranium reserves				
US FEA (1977) *	Cost	77 C	0.74	
US DOE (1987) *	Cost	83 C	3.08	

[a] Data definitions: C = cross section data. CT = cross section time series data. Diseq = estimated in a disequilibrium model. ns = reported as not significant with no value given. T = annual time series data. Tm = monthly time series data. Tq = quarterly time series data. SupplyDep = including depletion effect. SupplySys = systems approach. TX = Texas. * indicates the elasticity has been computed from information in the study. OH = Ohio, IL = Illinois, PA = Pennsylvania, KY = Kentucky, WV = West Virginia. N. App, C. App, S App = Northern, Central and Southern Appalachia.

markets. Green (1988) used full information maximum likelihood (FIML) estimation techniques on quarterly data for 1968:I to 1982:IV to model supply in the residual fuel oil market. Price is a function of residual fuel oil shipments, estimated value of the loss of an entitlement benefit from production, [2] interest rates, other petroleum product prices, refinery capacity, price of crude oil, price of imported fuel oil, price ceiling during periods of price control, lagged production of residual, lagged net imports, lagged residual fuel oil inventories, and lagged price. All product prices were measured as deviations from the price of crude oil. In this formulation the lagged production is presumably strongly related to current shipments and might represent expectations in the model. Inverting the elasticity point estimates from the price equation ($\partial \ln P/\partial \ln Q$) produced a short-run supply elasticity ($\partial \ln Q/\partial \ln P$) of 11.11 and a long-run elasticity of only 4.55. The inclusion of lagged price on the right-hand side of a price equation yielded this implausible result. [3]

[2] An entitlement benefit entitled oil buyers in the U.S. to purchase a certain percent of their consumption at the U.S. controlled price. However, the estimated loss of an entitlement benefit due to production is not explained in the paper, nor is it clear how the variable is defined.

[3] The author used a lagged price in the model to represent partial adjustment. To see why this formulation yields lower long-run than short-run elasticities consider the following simple model. Let $P_t = \alpha + \beta Q + \gamma P_{t-1}$. In this model, the first-period change in price from a change in quantity is $\mathrm{d}P/dQ = \beta$ reported by Green as 0.065 and the price elasticity with respect to quantity is β times Q/P which Green reported as 0.09. Since this is the inverse of the standard price elasticity of demand we get a short-run price elasticity of demand of $1/0.09$ or 11.11. To get long-run elasticities we must consider the changes over all periods. After one period the change $\mathrm{d}P_{t+1}/\mathrm{d}Q$ is $\beta\gamma$, after two periods it is $\beta\gamma^2$, etc. The total change $\mathrm{d}P/\mathrm{d}Q = \beta\Sigma\gamma^i$. If γ is less than 1 in absolute value (it equals 0.607 in the Green paper) then the total affect converges to $\beta/(1-\gamma)$. The long-run elasticity of price with respect to quantity is $\beta(Q/P)/(1-\gamma)$ which Green reports as 0.22. Inverting, we get a standard price elasticity of 4.55 which is smaller than the short-run elasticity.

Table 2
Secondary energy product supply elasticities for the United States [a]

Reference	Model type	Sample	Output elasticity		
			Short run		Long run
Residual fuel oil					
Green (1988) *	Supply	68:I–82:IV Tq	11.11		4.55
Rice and Smith (1977) *	Supply	46–73 T		−1.86	
Considine (1992) *	Cost	81:I–90:IV Tq	−28.57	4.00	4.00
Gasoline					
Koshal et al. (1991)	Supply	51–83 T	2.58		4.20
Tsurumi (1980)	Supply	70:1–76:12 Tm		0.53	
Tsurumi (1980)	Supply	70:1–76:12 Tm(shift)		1.96	
Tsurumi (1980)	Supply	70:1–73:10 Tm		1.98	
Tsurumi (1980)	Supply	74:3–76:12 Tm		0.77	
Yang and Hu (1984)	Supply	70:I–79:IV Tq		1.47	
Rice and Smith (1977) *	Supply	46–73 T		14.49	
Considine (1992) *	Cost	81:I–90:IV Tq	1.61	1.20	−100.00
Gasoline's share of oil product production					
Dahl (1981)	Supply	36–41, 47–75 T		0.20	
Dahl and Laumas (1981)	Supply	36–41, 47–76 T		0.08	
Considine (1992) *	Supply	81:I–90:IV Tq		0.57	
Jet fuel and kerosene					
Rice and Smith (1977) *	Supply	46–73 T		3.21	
Considine (1992) *	Cost	81:I–90:IV Tq	8.20	5.00	5.00
Distillate					
Rice and Smith (1977) *	Supply	46–73 T		9.71	
Considine (1992) *	Supply	81:I–90:IV Tq	1.57	1.18	1.61
Other oil products					
Considine (1992) *	Cost	81:I–90:IV Tq	−15.63	2.17	−1.67

[a] Data definitions: C = cross section data. CT = cross section time series data. Diseq = estimated in a disequilibrium model. ns = reported as not significant with no value given. T = annual time series data. Tm = monthly time series data. Tq = quarterly time series data. SupplyDep = including depletion effect. SupplySys = systems approach. TX = Texas. * indicates the elasticity has been computed from information in the study. OH = Ohio, IL = Illinois, PA = Pennsylvania, KY = Kentucky, WV = West Virginia. N. App, C. App, S App = Northern, Central and Southern Appalachia.

Koshal et al. (1991) estimated supply and demand equations for gasoline assuming that quantity supplied is a linear function of the price of gasoline, the price of crude oil, and input prices as measured by the producer price index. They used two stage least squares and found an own-price elasticity of 2.58 in the short run and 4.28 in the long run. A potential bias in this model arises from the omission of other product prices in the supply equation.

Dahl (1981) estimated a supply response function that measures the change in refineries' share of gasoline in the output mix to changes in product output prices.

She found a gasoline share elasticity of 0.20. Residual and distillate fuel oil are both substitutes for gasoline in production with distillate the better substitute.

Using a similar model and an additional year of data, Dahl and Laumas (1981) estimated the response of gasoline's share of refinery production to petroleum product prices. They found that the response of American refineries was stable and differed from refineries in Canada or the European Economic Community. For the U.S., they estimated the price elasticity of gasoline share to be 0.08 and the cross price share elasticities for residual and distillate fuel oil to be 0.07 and 0.11, respectively.

The problem of changes in the regression parameter estimates over time was addressed by Tsurumi (1980). He introduced a temporary shift function to analyze discontinuities in the U.S. gasoline market using monthly time series data from January 1970 to December 1976. He regressed the supply of gasoline on price, lagged inventories and binary variables for summer and winter. He found that the price elasticity of supply changed during the 1973–74 oil crisis from 1.98 before October 1973 to 0.77 after March 1974. His results show that treating two different regimes as one homogeneous group can lead to biased estimates of the elasticities. This same instability was not found by Dahl and Laumas (1981) when annual data and a longer time period were used to test for stability. Hence, the instability may reflect shorter term adjustment or may result from potential bias due to the omission of other product prices.

In addition to the equilibrium models reviewed above, disequilibrium models of energy product markets have also been used. Yang and Hu (1984) model the U.S. gasoline market using a disequilibrium approach, in which the quantity supplied is not always equal to quantity demanded, and the quantity exchanged is the minimum of the two. Yang and Hu's model consists of three equations. One equation describes the quantity traded; the other two describe supply and demand. In the supply equation, the quantity supplied of gasoline is a function of the price of gasoline, crude oil, residual fuel oil, distillate, and the volume of crude oil going to domestic refineries. From their estimates, they concluded that the price elasticity of gasoline supply is 1.47. In principle, this effort to model aggregate gasoline supply elasticities is the most pleasing theoretically. It includes the prices of other products, takes into account both supply and demand, as well as disequilibrium in the market. However, this model was estimated using quarterly data with no attempt to model seasonal effects on gasoline production. It seems unlikely that seasonal variation in prices would pick up all the seasonal effects, since Tsurumi found that summer and winter dummy variables were statistically significant in some of his regimes. Thus, the testing for seasonal variation in seasonal data seems warranted.

In their large U.S. petroleum market model, Rice and Smith (1977) included simple supply equations for gasoline, kerosene, distillate, and residual fuel. They modeled price as a function of quantity supplied, the average price of crude oil, and an undefined variable that appears to be some measure of the share of revenue from the product relative to the share of gasoline. The supply elasticities implied by their parameter estimates are 14.49 for gasoline, 3.21 for kerosene, 9.71 for distillate, and −1.86 for residual. The residual price elasticity, however, is not statistically significant. This type of econometric modeling may give rise to simultaneous systems bias with price being determined simultaneously by supply and demand. Omitted variables is also a potential problem. Dahl (1981), however, found that in a gasoline share equation, the price of products was exogenous suggesting the simultaneous systems bias may not be a problem. However, the production of each product is likely to be influenced by its own price as well as the price of other fuels. It is unclear whether the variable that represents the share of revenue relative to the share of gasoline is adequate to pick up the effect of these other prices.

To analyze the closely related market for natural gas, Barret (1992) used a disequilibrium model, an approach which he justified by arguing that the regulatory environment in the natural gas market caused the market to be out of equilibrium. He modeled the quantity exchanged as being equal to the minimum of quantity supplied or quantity demanded with quantity supplied a function of the price of natural gas and natural gas reserves. An equilibrium model from 1960–1990 yielded a supply elasticity of 0.014, disequilibrium models yielded somewhat higher elasticities varying from 0.02 to 0.15. Because reserves are included in his model his elasticities are best interpreted as short-run elasticities.

The supply of coal has also received attention by researchers. Labys et al. (1979) estimated the supply of steam coal, the total supply of coal, the surface supply of coal, and the underground supply of coal using ordinary least squares along with a number of other equations relating to the demand, supply, and stocks of coal. Before discussing this supply model, however, it is useful to have a bit of introductory information on the coal market. Coal can be divided into two categories by use: under the boiler use for process heat, which is called steam coal, and metallurgical use in the iron and steel industry, which is called metallurgical or coking coal.

Geologically coal is divided by characteristics into four major categories. Ranking them by their energy and carbon content they are anthracite, bituminous, subbituminous, and lignite. High carbon, high energy anthracite is relatively scarce in the US and is used mainly for thermal applications. All bituminous coal can be used as steam coal but only bituminous coals with special characteristics can be used for coking or metallurgical applications. Since metallurgical coals tend to be scarce relative to the demand for metallurgical coal, they are often too valuable to be used in the steam coal market and are usually used only in the metallurgical market. Further, since coking and noncoking coals are not found together, the price for steam coal, which is usually measured in dollars per million BTU, should determine the supply of steam coal to the market. Recently there has been crossover from the metallurgical to the thermal market suggesting that both metallurgical coal prices and steam coal prices might be relevant to steam coal supply. However, this crossover was minimal during the Labys study and its omission is not serious. The other two types of coal, both used in thermal applications, are subbituminous and lignite.

In their econometric work, Labys et al. (1979) estimated the steam coal market using annual U.S. data over the 1955–1973 time period. There is some ambiguity about whether the supply is for all four steam coals or just bituminous steam coals. Nor is it clear how the price is measured since no units are given, but from the numbers in simulations conducted, it appears to be in dollars per ton.

In the steam coal market, they found an elasticity of 0.61 in the short run and only 0.05 in a longer run adjustment of five years. This implausible result arises from using a lagged endogenous model with price as the endogenous variable. In such a model, if the coefficient on lagged price is between 0 and 1, which is likely, $\mathrm{d}P/\mathrm{d}Q$ is greater in the long run than in the short run implying that the elasticity is smaller in the long run than in the short run (see footnote 3 for an illustration of this point). If the goal is to distinguish between the long- and short-run elasticity of supply this formulation is inappropriate.

In their total coal supply, underground supply, and surface coal supply equations, quantity was estimated as a function of prices and lagged production. They found five-year supply elasticities of 0.05, 0.67, and 0.07 with long-run supplies of 0.11, 1.31, 1.00 for total, underground and surface coal supply, respectively. Although these estimates are more in line with theoretical expectations for the relationship between long- and short-run elasticities, they have two troublesome aspects. They unexpectedly suggest that the long-run supply of underground coal is more elastic than the long-run surface supply and that the total supply elasticity for coal is less elastic than either of its components.

For these total coal equations, the coal definition is somewhat ambiguous but appears to be all geological categories of coal given above adjusted into tons of coal equivalence. Price appears to be the price of bituminous steam coal per ton. Since metallurgical coal accounts for about a quarter of total coal consumption in their sample, a higher percentage than now, the omission of the price of metallurgical coal might cause their parameter estimates to be biased.

Although the above models do not suggest consensus point or even interval estimates for the various energy product elasticities, they do suggest a number of other conclusions. Simple models of oil supply perform poorly whether estimated alone or in conjunction with demand. Including depletion and price expectations tends to give results more in line with a priori expectations. Using a lagged endogenous model with price as the dependent variable does not produce satisfactory estimates of short- and long-run elasticities. There are large deviations in elasticities across various models for petroleum product models although most suggest rather elastic supply responses. Disequilibrium models appear to get larger elasticities than equilibrium models. The variability of results suggests that alternative modeling techniques be considered, which we do in the next section.

3. Supply elasticities from cost functions

A second approach to simple models of supply is to start with a cost function $C = C(Q)$. In a competitive world, the supply function is equal to the marginal cost function for the whole industry or $P = \partial C(Q)/\partial Q$. Solving for Q gives us the supply equation.

We will see two approaches to estimating cost or marginal cost functions in the literature. One way is to estimate the cost function directly as a function of output, the second is to compute reserve costs to derive reserve rather than supply elasticities.

The following studies use a variant of the first approach. Griffin and Jones (1986) estimated an operating cost function for Texas producing wells. Chermak and Patrick (1995) estimated an operating cost function for tight sands gas wells. Considine (1992) estimated a translog cost function for refinery operation. Zimmerman (1977) and Zimmerman (1981) (quoted from Kneese and Sweeney, 1993) estimated cost functions for coal. We will discuss each of these cost functions and relate them to the elasticities reported in Tables 1 and 2.

Griffin and Jones (1986) used a double log formulation and regressed operating costs using Texas well data for 1983 on the number of producing wells (W), lease oil production (Q), well depth ($Depth$), lease age (Age), ground water production (H_2O), plus three binary variables with non-defined oil lift mechanism ($D * pump$, $D * Pesp$, $D * Dgl$), the number of water injections wells (WH_2O) and total casing head gas (Gas). Their estimation results are [4]

$$\begin{aligned}\ln C = \underset{(2.67)}{2.85} + \underset{(5.98)}{0.53}\ln W + \underset{(3.76)}{0.28}\ln Q + \underset{(0.83)}{0.01}\ln Gas + \underset{(3.22)}{0.06}\ln H_2O \\ + \underset{(0.85)}{0.04}\ln WH_2O + \underset{(3.91)}{0.45}\ln Depth - \underset{(-1.22)}{0.09}\ Age + \\ + \underset{(3.30)}{0.82}D * Pump + \underset{(4.51)}{1.43}D * Pesp + \underset{(5.03)}{1.27}D * Dgl, \qquad R^2 = 0.897.\end{aligned}$$

Setting marginal cost from the above equation equal to price and solving for the price elasticity of production, we get $\partial \ln Q/\partial \ln P = -1.39$. [5]

Chermak and Patrick (1995) used monthly data from 1988 to 1990 on 29 different tight sand gas wells to derive an operating cost function for price taking extracting firms. They regress the log of production costs on the log of production

[4] The text claimed a log linear formulation but the equation did not specify a log form for age. We reproduce the equation as it was given in the original article, since it does not make any difference to the estimated supply elasticity included in Table 1.

[5] Since this and later estimates from cost functions are nonlinear combinations of the originally estimated parameters, they are consistent and asymptotically efficient provided the original estimators are at least consistent and asymptotically efficient.

(Q), the log of remaining recoverable reserves (R), the log of a time trend (*time*), and the log of a variable measuring how many months a well has been producing (*pm*). The result for their simplest model is

$$\ln C = \underset{(17.51)}{8.68} + \underset{(14.83)}{0.48} \ln Q - \underset{(-12.45)}{0.35} \ln R - \underset{(-6.83)}{0.50} \ln time + \underset{(6.78)}{0.13} \ln pm.$$

Computing marginal costs and setting them equal to price, we get a well supply elasticity of -1.92.

They reported results for five other equations that test for stability of the cost parameters across firms, geologic formations, locations, and wells. Although parameters vary with respect to these conditions, the results indicate that in all cases marginal costs decrease with extraction and increase with depletion. The implied price elasticities from these other experiments vary from -1.05 to -1.35.

Since both of the above studies use well data they are capturing the increasing returns to scale for wells that produce more output. They do not, however, capture aggregate supply elasticities as we aggregate over all wells over time. If we aggregated across wells at low prices only the low cost, high volume producers would remain in the market. As prices increase, higher cost, lower volume producers would find it profitable to remain in the market, thus tracing out an upward sloping aggregate supply curve. We also expect costs to go up over time as we deplete a well as is shown by the negative coefficient on lease age in Griffin and Jones or on remaining reserves in Chermak and Patrick.

Considine (1992) developed an interesting and theoretically-based model of product supply for gasoline (G), distillate (D), residual (R), jet fuel and kerosene together (J and K), and other products (OP). His 15-equation econometric model includes: (1) a restricted translog variable cost function with outputs of the five products, inventories for crude oil, work in progress, the five products, and the capital stock, (2) a cost share equation for energy and crude oil, (3–7) five revenue share equations for products, and (8–15) eight Euler equations– one for capital and seven for product inventories. His model was estimated using iterative three-stage least squares on monthly data for 1981:I to 1990:IV.

Considine's estimates include a short-run cost elasticity, or the percentage change in cost, which is price in a competitive market, with respect to a percentage change in output ($\partial \ln P/\partial \ln Q$) for each of the product categories. We report the more traditional inverse of these elasticities ($\partial \ln Q/\partial \ln P$). He found gasoline and distillate to have similar and statistically significant price elasticities both near 1.60. We have reported the estimated supply elasticities for the other products – residual, jet fuel and kerosene combined, and all other products – in Table 1. However, their $\partial \ln P/\partial \ln Q$'s are not statistically different from zero implying that their elasticities $\partial \ln Q/\partial \ln P$'s are infinite. Infinite elasticities might suggest that there is excess capacity for producing all products but gasoline and distillate, so we get little increase in cost when we produce more of them.

In his intermediate-run cost elasticities, he included changes in inventories in the adjustment process and these represent cost elasticities with respect to changes in shipments. Again inverting his cost elasticities, we find that supply price elasticities for gasoline and distillate are similar and near 1.20, while those for the other products are all above 2. In all cases the shipment elasticities are less elastic than the production elasticities suggesting that cost of product out of inventories is greater than the direct cost of production. It also suggests to us that these shipment elasticities might be more appropriately interpreted as short-run elasticities, while the production elasticities, which are more elastic, might better be interpreted as intermediate-run elasticities.

In the long run, he also allowed the capital stock to adjust. He found the long-run elasticity for gasoline to be -100. This very elastic response shows a slight decrease in costs as shipments increase which might reflect increasing returns to scale in refinery production of gasoline. We see a more pronounced downward sloping supply with a price elasticity of -1.67 for other products suggesting that economies of scale or perhaps technological change has decreased costs by reducing required capital as output has gone up. For residual and jet fuel, the long-run elasticities at 4 and 5, respectively, are similar to those for the short run, while the distillate elasticity at 1.61 is a bit more elastic than in the short run.

We can relate these short-run gasoline supply elasticities to the share elasticities in Dahl (1981) above. We know that

$$\ln(G/O) = \ln G - \ln O,$$

yielding a share elasticity of

$$\partial \ln(G/O)/\partial \ln P_g = \partial \ln G/\partial \ln P_g - \partial \ln O/\partial \ln P_g.$$

We can compute this elasticity from the short-run elasticities in Considine as follows:

$$\begin{aligned} &= \partial \ln G/\partial \ln P_g - \partial \ln O/\partial \ln G * \partial \ln G/\partial \ln P_g \\ &= 1.605 - 0.647 * 1.605 = 0.567. \end{aligned}$$

This is somewhat more elastic than the elasticity in Dahl (1981) of 0.20 but given the changes in the energy market, we might expect somewhat more flexibility as the small tea kettle refineries that sprouted up with the oil quota have disappeared and we are left with more sophisticated refineries.

The last study that has estimates of operating costs is Zimmerman (1977). He estimated production per section as a function of seam thickness and number of machines working. He used this production function combined with estimates of cost to derive total cost as a function of seam thickness and output for deep mined coal using mine data from Ohio, Illinois, Pennsylvania, Kentucky, and W. Virginia. Although the exact date is not specified, data appear to be taken from publications in 1973.

The resulting cost functions in the 1977 paper for drift and deep mines are:

$$\text{drift:} \quad TC = 1{,}365{,}256 + 1{,}655{,}781\left[Q/\left(684Th^{1.42174}\epsilon\right)\right]^{1.498},$$

$$\text{deep:} \quad TC = 2{,}355{,}999 + 1{,}655{,}781\left[Q/\left(684Th^{1.42174}\epsilon\right)\right]^{1.498},$$

where Th is the seam thickness and ϵ is a variable that measures the random geological characteristics of the seam. Differentiating the total cost functions with respect to quantity and setting marginal cost equal to price yields an implied elasticity of 2.03 for both type of mines.

Zimmerman (1981) used the same technique and found a cost function for deep mined coal as

$$\text{deep:} \quad TC = 3914764 + 2122480\left[Q^{1.2796}\left(1597.6Th^{1.1071}\right)\epsilon\right],$$

which yields a price elasticity of 3.58.

A second approach to estimating a cost function for remaining reserves is to compute costs of producing various reserves as in Dahl (1992) for gas, US FEA (1977) and US DOE (1987) for uranium, and Lin (1978) for coal. We use the reserve cost estimates from these studies and line them up with reserves to get an elasticity of reserves with respect to price rather than output with respect to price.

Dahl (1992) computed gas reserve costs by state using drilling costs and new reserve additions. By lining up these reserves and cost, we derive the following U.S. reserve function for natural gas:

$$\ln R = \underset{(50.832)}{5.828} + \underset{(9.958)}{0.405} \ln P, \qquad R^2 = 0.917,$$

yielding a gas reserve elasticity of 0.40.

The US FEA (1977) and US DOE (1987) estimated the amount of reserves of uranium that would be economic at various uranium prices. Using their values for 1977, we regressed the log of reserves on prices ranging from \$10 to \$50 to get the following reserve function:

$$\ln R = \underset{(33.29)}{10.82} + \underset{(7.19)}{0.74} \ln P, \qquad R^2 = 0.86,$$

yielding a reserve elasticity of 0.74. The US DOE (1987) updated their model for 1983 data and again estimated the amount of reserves that would be forthcoming at uranium prices from \$22.50 to \$45.00 per pound. Our regression for this updated data is

$$\ln R = \underset{(-1.84)}{-4.53} + \underset{(4.26)}{3.08} \ln P, \qquad R^2 = 0.86,$$

yielding a more elastic reserve elasticity of 3.08.

Lin (1978) computed surface mining reserve costs for Northern Appalachia, Central Appalachia, and Southern Appalachia under three different regimes: zero backfill, 50% backfill, and 100% backfill. His costs are based on a process model for 1972 production. Matching up his reserves with his marginal costs, we find implied reserve elasticities that vary from 0.41 to 0.79 for Northern Appalachia, from 7.47 to 7.9 for Central Appalachia, and from 0.69 to 0.81 for southern Appalachia.

All these reserve cost studies obtained reasonable results for supply elasticities when applied to coal, natural gas, and uranium. Further, such studies require neither perfect competition nor static optimization by producers. Therefore, these cost functions can be used in simulation models that assume a more complex market structure. For these reasons, it seems that reserve cost modeling is a productive approach and we apply it to oil in the next section.

4. U.S. oil reserve supply function

We begin by modeling oil reserve costs for the U.S. using reserve cost information computed in Dahl (1990). She classified reserves by the three categories employed in the U.S. Geological Survey. The data are summarized in Table 3 by U.S. region. Proved or measured reserves in 1987 are 29.5 billion barrels, indicated and inferred reserves are 21.7, and undiscovered reserves are 49.4 giving a total of 100.6 billion barrels for all three reserve categories.

In the next step, she developed cost estimates for each reserve category. For developed reserves the cost of production is lifting costs, which include all variable or operating costs except taxes. They do not include any exploration or development costs. To determine how many reserves have been developed in the US, we begin by estimating the age of producing wells in the US using the drilling history of oil wells and the number of producing wells from 1859 to 1988 from API statistics. [6]

The number of abandonments (WA_i) equals producing wells last year (WP_{i-1}) plus wells drilled this year (WD_i) minus producing wells this year (WP_i) or

$$WA_i = WP_{i-1} + WD_i - WP_i.$$

Attributing abandonments in each year to the oldest wells yields the 1988 vintage structure of wells listed in column 1 of Table 4.

For example, these computations suggest that in 1988 there were 16738 wells left that were drilled in 1960 whereas the number of 1988 wells was only 13057, reflecting much lower drilling in that year. Taking a vintage weighted average yields the average well age in 1988 as 12.43 years.

To determine how many reserves each of these wells represents, we begin by determining how many reserves each well originally represented by dividing new

[6] To the best of our knowledge the vintage structure of wells is not readily available.

Table 3
U.S. Oil: Costs, production and reserves 1986/1987. Cost in \$/barrel, reserves in billions of barrels (bb)

	$\text{Cost}_{e\&d}$/barrel 1986/87	Reserves			
		Meas bb	Inf & In bb	Und bb	(2+3+4) bb
	(1)	(2)	(3)	(4)	(5)
Federal Offshore				3.4	3.4
Alaska Offshore	-	-	-	3.4	4.9
Pacific Offshore	\$4.00	1.3	0.2	8.6	13.0
Gulf Offshore	\$3.78	4.0	0.4	0.7	0.7
Atlantic Offshore	-	-	-		
State Onshore and Offshore					
Alaska	\$1.27	6.9	6.4	13.2	26.5
Pacific	\$2.69	4.7	1.2	3.5	9.4
Colorado Plateau and Basin and Mountain Range	\$3.93	0.6	0.4	1.5	2.6
Rocky Mountains and Northern Great Plains	\$14.43	1.2	1.3	4.5	7.1
West Texas and	\$11.17	5.4	3.8	2.6	11.9
Gulf	\$9.35	3.7	5.7	4.2	13.6
Mid-continent	\$13.72	1.1	1.4	1.9	4.4
Eastern Interior	\$13.06	0.5	0.7	1.8	3.0
Atlantic Coast	\$10.27	0.0	0.0	0.2	0.3
US Total	\$6.40	29.5	21.7	49.4	100.6

Source: Dahl (1990, p. 30).

reserves in year v, shown in column (2), by the number of producing wells of vintage v in column (1). The resulting $(R/W)_v$ is shown in column (3).

Examining these numbers, we can see the effect of Alaska North slope discoveries with large new vintages of oil in 1970. We can also see the year to year variations in new reserves per well reflecting changes in drilling patterns and discoveries.

Choosing a decline rate of 0.101, which equals 1988 production over reserves, 1988 production for each well of vintage v is given in column (4) and is computed by

$$(Q/W)_v = ((0.101)(R/W)_v \, e^{(-0.101(1988-v))}).$$

Total production for each vintage given in column (5) is equal to column (4) production per well of each vintage times the number of wells of each vintage in column (1). Summing production over all vintages gives us an estimate of total production of 2.76 billion, which is rather close to the actual production of just under 3 billion barrels in 1988.

To determine how many developed reserves each of these wells of vintage v

Table 4
1988 wells, production, and reserves by well vintage

Vintage year	(1) Wells by vintage in 1988	(2) New reserves oil (10^3 bbl)	(3) New reserves/well (10^3 bbl)	(4) Production/well (10^3 bbl)	(5) Production by vintage	(6) Remaining reserves/well (10^3 bbl)	(7) Remaining reserves by vintage
1960	16738	2468797	147.50	0.88	14744	8.72	145982
1961	21101	2767052	131.13	0.87	18282	8.58	181007
1962	21249	2306907	108.57	0.79	16861	7.86	166945
1963	20288	2333490	115.02	0.93	18868	9.21	186815
1964	20620	2807342	136.15	1.22	25112	12.06	248637
1965	18761	3210395	171.12	1.69	31770	16.77	314552
1966	16780	3127499	186.38	2.04	34239	20.20	338996
1967	15329	3140285	204.86	2.48	38032	24.56	376557
1968	14227	2659489	186.93	2.50	35632	24.80	352796
1969	14368	2297496	159.90	2.37	34054	23.47	337166
1970	13043	12885923	987.96	16.20	211295	160.39	2092031
1971	11903	2515536	211.34	3.83	45632	37.96	451801
1972	11437	1731819	151.42	3.04	34754	30.09	344098
1973	10251	2321334	226.45	5.03	51535	49.78	510248
1974	13664	2152702	157.55	3.87	52870	38.31	523470
1975	16979	1488950	87.69	2.38	40455	23.59	400545
1976	17697	3796053	214.50	6.45	114101	63.84	1129711
1977	18700	1287265	68.84	2.29	42804	22.66	423806
1978	19065	2750927	144.29	5.31	101196	52.55	1001940
1979	20689	1576310	76.19	3.10	64149	30.70	635138
1980	32120	3141365	97.80	4.40	141426	43.59	1400259
1981	42520	2749624	64.67	3.22	136946	31.89	1355898
1982	39252	1588715	40.47	2.23	87536	22.08	866690
1983	37396	3047999	81.51	4.97	185788	49.19	1839484
1984	44472	3960696	89.06	6.01	267078	59.46	2644335
1985	36458	3244553	88.99	6.64	242039	65.73	2396424
1986	18537	1641252	88.54	7.31	135447	72.34	1341059
1987	15787	3414378	216.28	19.75	311723	195.50	3086369
1988	13057	2218750	169.93	17.16	224094	169.93	2218750
Total	612488			Forecast	2758462	Forecast	27311509
Av Age	12.43 years			Actual Q	2979123	Actual R	27256000

Source: Computed from information in the American Petroleum Institutes *Basic Petroleum Data Book* (1990) and *Petroleum Facts and Figures* (1971).

represents in 1988 $(R/W)_{v1988}$ (Column 6), we take original reserves per well and subtract off cumulative production or

$$(R/W)_{v1988} = (R/W)_v - \int_0^{1988-v} \left[0.101(R/W)_v e^{(-0.101(x))}\right] \mathrm{d}x.$$

Total developed remaining reserves by vintage in column (7) is column (6) times wells per vintage in column (1). Total developed reserves is the sum of developed reserves per vintage over all vintages, which is equal to about 27.3 billion barrels. This number, which is rather close to proven reserves for 1988, suggests that most proven reserves are also developed reserves. Hence, we will take proven reserves as a measure of developed reserves for the U.S. For these reserves, costs of production will be taken as lifting costs. [7]

Lifting costs without taxes of \$4.64 in 1987 are taken from Dahl (1990, p. 50). They were computed by taking total lifting costs and distributing them over oil and gas according to lifting costs of typical wells taken from *Oil and Gas Journal*, 2/20/89:49.

$$C_{\text{lift}} = TC_{\text{lift}} / \left(^{\Delta} RO + @^{\Delta} RG\right).$$

Using these weights and measuring oil in barrels and gas in MCF gives $@ = 0.177$ times \$0.8/\$2.82 = 0.05. Above ground costs (P_p) which include a normal rate of return r are $P_p = C_p(1 + r)$.

Converting \$4.64 into above ground costs at an 8% real interest rate gives lifting costs of \$5.01 which will be taken as the cost of producing the 29.5 billion barrels of proven reserves given in Table 3.

A second category of reserves shown in Table 3 is indicated and inferred reserves of 21.7 billion barrels. Costs for this category of reserve are taken to be the costs of secondary recovery from Roumasset et al. (1983). They estimated these costs to be between \$2 and \$10 in 1982 dollars. Changing them into 1987 dollars would yield costs from \$2.36 to \$11.77. Linearizing gives the cost function in terms of remaining indicated and inferred reserves ($Rm_{i\&i}$) as

$$C = 11.77 - 0.000443 Rm_{i\&i}$$

with costs in 1987 costs per barrel and reserves in millions of barrels.

The last category of reserves are the 49.4 that are undiscovered. The exploration and development costs for these are taken from Dahl (1990). To compute these values Dahl applied a technique by Adelman and Shahi (1989) using cost information from the *Joint Survey of Drilling*. In this technique exploration and development cost of a barrel of oil in the ground ($C_{e\&d}$) are costs per well (C_w) times number of oil wells drilled (W_o) + the share of dry wells attributed to oil

[7] This same computation was also done for each year between 1958–1987. The average difference between estimated and actual production was 5% and the average difference between estimated and proven reserves was 7% again supporting the supposition that proven reserves are essentially developed reserves.

(δW_d) times 1.66 to include all non drilling exploration and development expenses, all divided by reserve additions for oil or

$$C_{e\&d} = C_w(W_o + \delta W_d)1.66/{}^{\Delta}R.$$

Costs summarized in Table 3 are the averages for 1986/1987 aggregated from state and state regional data for some of the larger producing states. The more disaggregated numbers are also given in the original source.

The current finding cost structure of the US by region in 1986/87 is converted to above ground costs by $\$ = (C_{e\&d})(\alpha + r)/\alpha$, where $C_{e\&d}$ is the in ground costs per barrel given in Table 3, α is the decline rate assumed to be 0.1 and r is the real interest rate assumed to be 8%.

Ordering all three sets of reserves by costs and assuming that marginal costs equals price, we get a reserve supply function with reserves in millions of barrels and price in 1987 $ per barrel:

$$\ln R = \underset{(16.75)}{7.44} + \underset{(5.48)}{1.27} \ln P, \qquad R^2 = 0.73,$$

giving an oil reserve elasticity of 1.27. The results from this approach appear quite reasonable. Compared to other products, oil supply appears more elastic than natural gas, but less elastic than coal. Because the estimates are based on cross sectional estimates, they do not include any technical change suggesting all these elasticities would have to be adjusted over time to reflect technical progress in each of the markets. Nevertheless, the reserve cost approach appears to be a promising for estimating energy supply elasticities.

5. Conclusions

Our goal in this paper has been to determine whether energy producer behavior can be reliably represented by price elasticities and to apply the most promising approach to the oil market. We began our investigation with a survey of studies using simple models which have directly estimated price elasticities or that contain cost functions from which supply price or reserve elasticities can be computed.

Simple models of aggregate oil supply perform badly and tend to get negative or statistically insignificant supply elasticities. [8] Results improve when expectations and reserve depletion are included but not when a simultaneous system is estimated. The one reasonable estimate yields a price elasticity of 0.58.

Models that represent supply with price as the dependent variable with a lagged endogenous variable on the right hand side of the equation do not capture the

[8] We base our conclusions on a priori expectations, which is clearly one out of a number of criteria that can be employed. For those readers with different a priori expectations, we hope we have provided enough model information for them to come to their own conclusions.

correct distinction between long and short-run response. Disequilibrium models appear to get a more elastic price response than equilibrium models.

Where operating costs have been estimated for oil or natural gas on a well basis they imply negative supply elasticities, which vary from −1.05 to −1.92, reflecting economies of scale. We do not feel these represent long-run aggregate elasticities because at lower prices only the cheaper wells with more production per well would remain in the market. However, as prices rise less efficient wells would engage in production, tracing out an upward sloping aggregate supply curve.

Petroleum products have been modeled in a variety of ways including either supply or cost estimation. We see considerable variation across the estimates with supply elasticities varying from −100 to 14.49 for gasoline, 1.18 to 9.71 for distillate; 3.21 to 8.20 for jet fuel and kerosene, −28.57 to 11.11 for residual, and −15.63 to 2.17 for other oil products. From these estimates it is hard to pin down the supply elasticity of oil products, but all approaches suggest they are rather elastic.

In general, more sophisticated models estimate a more elastic supply response. A systematic study of refinery behavior comparing modeling approaches would be useful, as would more investigation whether the response has been stable over time.

Models that line up computed costs with reserves to yield reserve price elasticities all give plausible results. They find a gas reserve elasticity of 0.41, uranium reserve elasticities varying from 0.74 to 3.08 and Appalachian coal reserve elasticities varying from 0.41 to 7.90.

Among the simple modeling methods, this last approach appears to have the most promise for primary energy supply, and we have applied it to the U.S. oil market. Costs have been developed for proven, inferred and indicated, and undiscovered reserves. By lining up reserves with costs, we have derived a reserve supply function for the US that yields an oil reserve elasticity of 1.27.

Our results are based on costs for a short time period. It would be useful to apply the same analysis over a long period of time to capture the affects of depletion and technical change and to determine how robust this cost function is over time.

These reserve equations are estimated from costs rather than from market behavior and are therefore more general in nature, as they assume neither competitive nor statically optimizing behavior. For this reason they can also be used in simulation models to investigate whether competitive market conditions and rationality appear to hold. Although data may be hard to acquire, our survey work suggests that for a better understanding of primary energy supply, it is worth the extra effort to develop and apply such data.

We also encourage work in areas of secondary energy supply, including electricity and renewable energy sources, for which we have found no simple supply models.

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